



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B.Sc Chemistry				
Course Title	CHEMISTRY OF CONSUMER PRODUCTS				
Type of Course	VALUE ADDED COURSE (VAC)				
Semester	III				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	Fundamentals of organic chemistry Foundations of analytical chemistry				
Course Summary	This course delves into the scientific principles behind everyday items such as soaps, detergents, shampoos, and cosmetics. Students learn about the chemistry of manufacturing, formulation techniques, and quality control procedures. Topics include how ingredients like linear alkyl benzene and sodium lauryl sulfate are synthesized, as well as the creation of specialized products like anti-dandruff shampoos and herbal soaps. Environmental impact and regulatory compliance are also covered. Through theory and practical lab work, students gain a solid understanding of the chemistry driving these commonly used consumer goods.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the process of making soaps from oils and fats, including the formulation of different types like herbal and medicated soaps.	U	C	Instructor-created exams / Quiz
CO2	Identify the ingredients and functions used in detergent production, comparing their effectiveness with traditional soap.	U	F	Class test /Assignment / Quiz
CO3	Understand the components in anti-dandruff and herbal shampoo, and their safety standards.	U	F	Class test /Assignment / Quiz

CO4	Analyze cosmetic preparation ingredients and functions to ensure the production of safe and effective products like face creams and nail polishes.	An	C	Class test /Assignment / Quiz
CO5	Evaluate the environmental impact of consumer products, proposing sustainable practices to minimize harm.	E	C	Class test /Assignment / Quiz
CO6	Analyze the need for innovative solutions to challenges in consumer product chemistry, fostering creativity and improvement in the industry.	An	F	Class test /Assignment / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hrs (45)	Marks
I		Soaps	9	18
	1	Saponification of oils and fats. Manufacture of soaps.	1	
	2	Formulation of toilet soaps. Different ingredients used and their functions.	2	
	3	Medicated soaps. Herbal soaps. Mechanism of action of soap.	2	
	4	Soft soaps. Shaving soaps and creams.	2	
	5	ISI specifications of soaps and creams. Testing procedures/limits.	2	
II		Detergents	9	18
	6	Types of Detergents. Anionic detergents: Manufacture of LAB (linear alkyl benzene). Sulphonation of LAB – preparation of acid slurry.	1	
	7	Different ingredients in the formulation of detergent powders and soaps. Liquid detergents. Foam boosters. AOS (Alpha Olefin Sulphonates)	2	
	8	Cationic detergents: examples. Manufacture and application. Non-ionic detergents: examples. Manufacture of ethylene oxide condensate.	2	
	9	Mechanism of action of detergents. Comparison of soaps and detergents.	2	
	10	Biodegradation – environmental effects. ISI specifications / limits for detergents.	2	
III		Shampoos	8	15
	11	Manufacture of SLS and SLES. Ingredients. Functions.	2	
	12	Different kinds of shampoos – anti-dandruff, anti-lice, herbal and baby shampoos.	2	

	13	Hair dye. Manufacture of conditioners. Coco betaines or coco diethanolamides	2	
	14	ISI specifications for shampoos. Testing procedures and limits.	2	
		Cosmetic Preparations	10	19
IV	15	Face and skin powders. Ingredients, functions. Different types.	2	
	16	Snows and face creams. Chemical ingredients used. Anti perspirants. Sun screen preparations.	2	
	17	UV absorbers. Skin bleaching agents. Depilatories. Turmeric and Neem preparations. Vitamin oil.	2	
	18	Nail polishes: nail polish preparation, nail polish removers. Article removers.	2	
	19	Lipsticks, roughes, eyebrow pencils. Ingredients and functions. Hazards of cosmetic preparations. ISI specifications.	2	
IV	UNI T- V	Open Ended: Leading firms, brand names, choosing the right product. Packing regulations. Marketing. Licensing – drug license – legal aspects. GMP – ISO 9000/12000 – consumer education. Evaluation of the product – advertisements. Visit to a cosmetic production facility	9	

References

1. Gobala Rao.S , Outlines of chemical technology, Affiliated East West press,1998
2. Kafaro, Wasteless chemical processing, Mir publishers, 1995.
3. Sawyer.W, Experimental cosmetics,Dover publishers, New york, 2000.
4. Ayaz Mahmood Dar, Cosmetic Chemistry: An Instant Approach. Educreation Publishing, 2011.
5. P. K. Chattopadhyay, Modern Technology of Soaps, Detergents & Toiletries (with Formulae & Project Profiles) 4th Revised Edition, NIIR Board publication.

Mapping of COs with PSOs and Pos

	PS O1	PS O2	PS O3	PS O4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1		1			2	1					1	2
CO 2	1		1			2	1					1	2
CO 3	1		1			2	1					1	2
CO 4	1		2			2	1					1	2
CO	1		1			2	1					2	2

5													
CO 6	1		2	1		2	1					2	2

Correlation Levels :

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory	Assignment / Viva	End Semester Examination
CO1	✓	✓	✓
CO2	✓	✓	✓
CO3	✓	✓	✓
CO4	✓	✓	✓
CO5	✓	✓	✓
CO6	✓	✓	✓